Dynamic Web Pages with Selenium

Week 8 · Documents module · Textbook Chapter 8

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This week at a glance

Most of the modern web renders its data with JavaScript *after* the HTML arrives. This week we drive a real browser to reach it.

Monday | Concepts & notebook intro

- Why JavaScript content is invisible to `requests`; Selenium as a browser you script

Wednesday | Notebook lab

- Work the Chapter 8 notebook: xkcd, a Wikipedia search, and an infinite-scroll page

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 8

Companion notebook

`ch-08-dynamic-pages.ipynb`

Bring a laptop

Wednesday is hands-on — a browser will open and click by itself.

1. Monday · Concepts

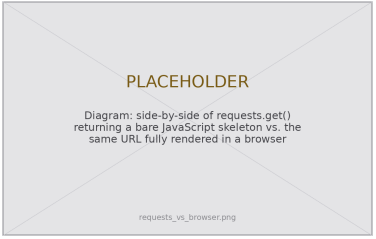
When static scraping fails

Last week's tools — `requests` + BeautifulSoup — work on **static** pages, whose content is fully present in the initial HTML.

But most modern sites are **dynamic**: their content is loaded or rendered by JavaScript *after* the HTML arrives. `requests.get()` hands you the skeleton *before* that code has run.

The data you see in your browser may simply not exist in the response. Diagnose it: compare `requests.get(url).text` to the page in your browser — if the content you want is missing, the page is dynamic.

The fix: a tool that *executes* JavaScript — a real browser, driven by code. That tool is **Selenium**.



Same URL, two very different payloads.

Textbook Ch. 8, "When Static Scraping Fails."

Selenium: a browser you drive with code

Selenium needs two things: the `selenium` library and a **browser driver** — a small program that lets Python control a specific browser.

Since v4.6 (late 2022), **Selenium Manager** finds or downloads the right driver automatically, so in practice you only `pip install selenium`. Creating a driver is a one-liner:

- `driver = webdriver.Chrome()` opens a real window
- every command you issue through `driver` happens *in that window*

You are not faking HTTP requests any more — you are **automating a browser**, the same one a person would use.

"Controlled by automated test software."

Textbook Ch. 8, "Setting Up Selenium."

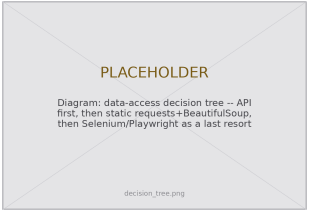
A powerful last resort — and not a loophole

Selenium is slow, resource-intensive, and brittle. Use it only when the simpler tools cannot. The decision tree:

1. **API?** Use it — fastest, most structured.
2. **In the static HTML?** `requests` + BeautifulSoup.
3. **Rendered by JavaScript?** *Now* reach for a browser.

Playwright (Microsoft) is the modern alternative — auto-waiting, faster, async — but Selenium is more widespread and fine for coursework.

Researchers increasingly *need* automation as platforms enclose once-open data (Freelon 2018) — which raises the stakes, not lowers them.



Automation is not a loophole

A `robots.txt` rule and a site's ToS still apply when a real browser does the clicking. (Ch. 2)

Textbook Ch. 8, "When to Use Selenium vs. APIs" & "The Fragility Problem."

What the notebook covers

This week's companion notebook builds the workflow one interaction at a time:

- **Launch & locate** — start a driver, find elements by ID, name, CSS selector, and XPath (on xkcd)
- **Extract** — read hidden attributes like an image's `alt` and `title` (the xkcd hover-text joke)
- **Interact** — click a button, type a Wikipedia search, and submit it like a user
- **Wait & scroll** — replace blind `time.sleep()` with `WebDriverWait`; harvest an infinite-scroll page
- **Hand off** — pass the rendered `page_source` to BeautifulSoup, then `driver.quit()`

Connecting to Ch. 2 (Ethics)

Everything from the ethics chapter applies with full force here. Before you automate *any* site, check its `robots.txt` and Terms of Service — especially anything logged-in or ToS-restricted, where the temptation is greatest.

2. Wednesday · Notebook Lab

Launch a driver, then locate elements

Four strategies find elements; `find_element` returns the first match, `find_elements` returns a list:

```
from selenium import webdriver
from selenium.webdriver.common.by import By

driver = webdriver.Chrome() # Selenium Manager fetches the driver
driver.get("https://xkcd.com")

comic = driver.find_element(By.ID, "comic") # by id
nav = driver.find_elements(By.CSS_SELECTOR, "ul.comicNav li a") # -> a list
title = driver.find_element(By.XPATH, "//div[@id='ctitle']") # by XPath
print(title.text) # the title of the current comic
```

Prefer a stable anchor — an `id` or a semantic `class` — over a brittle, position-based path.

Textbook Ch. 8, “Navigating and Finding Elements.”

Read attributes, then act like a user

```
# xkcd hides a joke in the image's title (hover) attribute
img = driver.find_element(By.XPATH, "//div[@id='comic']//img")
print(img.get_attribute("alt")) # the accessible description
print(img.get_attribute("title")) # the mouseover punchline

# Anything a user can do, Selenium can do -- e.g. click "Random":
btn = driver.find_element(
    By.XPATH, "//ul[@class='comicNav']//a[contains(text(),'Random')]")
btn.click()
```

`get_attribute()` reads *any* HTML attribute (even ones you never see); `.text` reads the visible text.

Textbook Ch. 8, “Extracting Data” & “Simulating Interactions.”

Type, submit — then wait *well*

Fill a search box, submit, and wait for the exact condition instead of guessing a duration:

```
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver.get("https://www.wikipedia.org")
box = driver.find_element(By.NAME, "search")
box.send_keys("Colorado Buffaloes") # type like a user
box.send_keys(Keys.RETURN) # submit the form

WebDriverWait(driver, 10).until( # not a blind time.sleep(2)!
    EC.presence_of_element_located((By.ID, "firstHeading")))
print(driver.title) # -> Colorado Buffaloes - Wikipedia
```

`WebDriverWait` polls every 0.5 s, returns *the instant* the condition holds, and raises `TimeoutException` if it never does — faster, and it fails loudly. Common conditions: `presence_of_element_located`, `visibility_of_element_located`, `element_to_be_clickable`.

Textbook Ch. 8, "Typing and Searching" & "Waiting for Dynamic Content."

Scraping an infinite-scroll page

Scroll to the bottom, wait, and stop when the page stops growing:

```
def scrape_infinite_scroll(driver, max_scrolls=20, pause=2):
    last = driver.execute_script("return document.body.scrollHeight")
    for i in range(max_scrolls):
        driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
        time.sleep(pause) # let new content load
        new = driver.execute_script("return document.body.scrollHeight")
        if new == last: # height stopped growing -> the end
            break
        last = new
    return driver.page_source # the fully-loaded HTML
```

Set a sane `max_scrolls`. **Proportionality** (Ch. 2): collect only what your question needs — not an entire 10,000-post feed.

Textbook Ch. 8, “Scraping Infinite Scroll Pages.”

Selenium renders; BeautifulSoup reads

Get the page into the state you need, hand the rendered HTML to BeautifulSoup, and **always** clean up:

```
from bs4 import BeautifulSoup

try:
    driver.get("https://example-search-site.com")
    # ...scroll / click / search until the page is ready...
    soup = BeautifulSoup(driver.page_source, "html.parser") # rendered HTML
    for item in soup.find_all("div", class_="result-item"):
        title = item.find("h3")
        if title is None: # same defensive habit as static scraping
            continue # skip a malformed item instead of crashing
        print(title.get_text(strip=True))
finally:
    driver.quit() # each driver is a whole browser process
```

Selenium is the **hands** that navigate; BeautifulSoup is the **eyes** that read. `try/finally` guarantees the browser closes even when parsing throws.

Textbook Ch. 8, "Passing to BeautifulSoup" & "A Practical Selenium Workflow."

Lab: work these, then show-and-tell

Work in pairs. Do these exercises from the chapter:

1. Extract data that JavaScript loads. Compare it with the data that `requests.get()` returns.
2. Automate a search that has more than one step. Navigate to the page, enter the query, submit it, and extract the result.
3. Examine three sites. Categorize each site as static, partially dynamic, or fully dynamic.
4. Find a "Load More" button. Click it in a loop until it disappears. Count the items that were hidden.
5. Measure the time for `requests`, for Selenium with a window, and for Selenium with no window. Report when the slower method is acceptable.
6. **5617**: use both methods on a research site that depends on JavaScript. Write a memo of about 500 words. Refer to Mitchell (2018)

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

The Twitter scraper that worked in 2019 was dead by 2024.

Browser automation is borrowed time — write it defensively, and expect it to break.

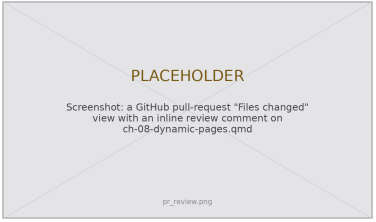
3. Friday · Textbook Revisions

Improving Chapter 8 together

Today we make the book better. Each of you opens a **pull request** against `ch-08-dynamic-pages.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 8

High-value targets — pick one and scope it to a single PR:

- **Test the code against live sites.** The xkcd and Wikipedia examples drift. Run the notebook headless, report what broke (a renamed class, a new consent banner), and fix the selector — the chapter's own "Fragility Problem."
- **Close a gap in "Common Issues to Debug."** Add a worked `StaleElementReferenceException` reproduction, or a headed-vs-headless content difference caused by `navigator.webdriver` detection.
- **Write the Playwright counterpart.** The chapter *names* Playwright as the modern alternative but shows no code — add a short side-by-side of the search example.
- **Add a figure.** A clean API → static → Selenium decision-tree diagram, or an annotated shot of a driver-controlled browser.
- **Strengthen an exercise.** Add expected output, a rubric, or a starter cell (e.g. a public infinite-scroll demo page) so Exercise 1 or 4 is self-checking.
- **Extend "Further Reading."** A public-interest case study that turned to browser automation after an API closed — tying the "Public Interest" callout to the post-API enclosure.

Targets drawn from the chapter's callouts, "Common Issues," and "Further Reading."

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Extracting data from PDFs —** when the data isn't on a web page at all.

Key takeaways

1. Static tools see the HTML **skeleton**; dynamic pages render their data with JavaScript *after* it arrives.
2. Selenium drives a **real browser —** locate elements (ID / name / CSS / XPath), then click, type, and scroll like a user.
3. Wait on **conditions** with `WebDriverWait`, not a blind `time.sleep()` — it's faster and fails loudly.
4. Render with Selenium, extract with BeautifulSoup — hands and eyes — and `driver.quit()` every time.
5. Selenium is a slow, fragile **last resort**: API → static → browser. Automation never suspends `robots.txt` or a site's ToS.

Reading: Textbook Ch. 8. Related: Mitchell (2018); García et al. (2020); Perriam et al. (2020).